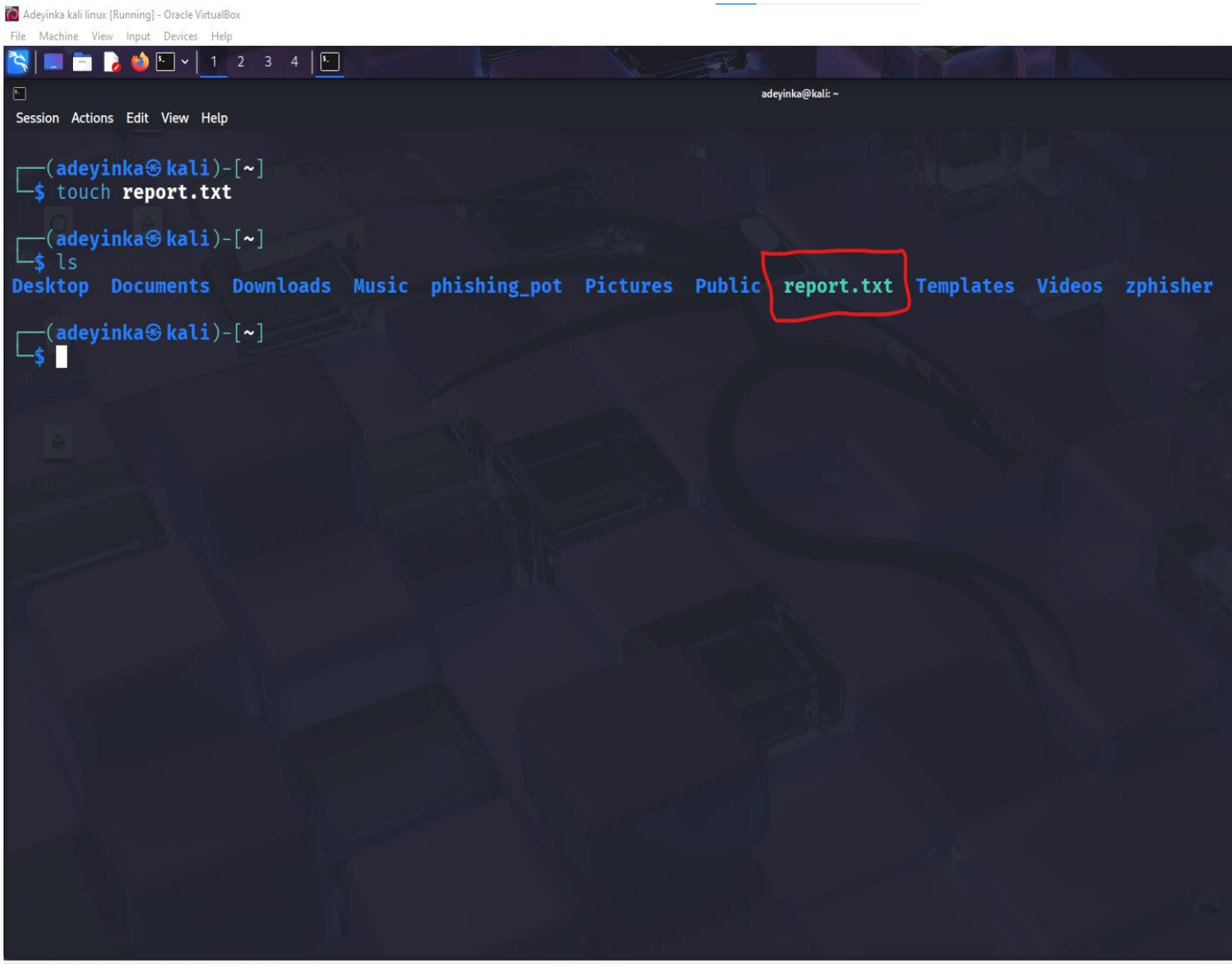
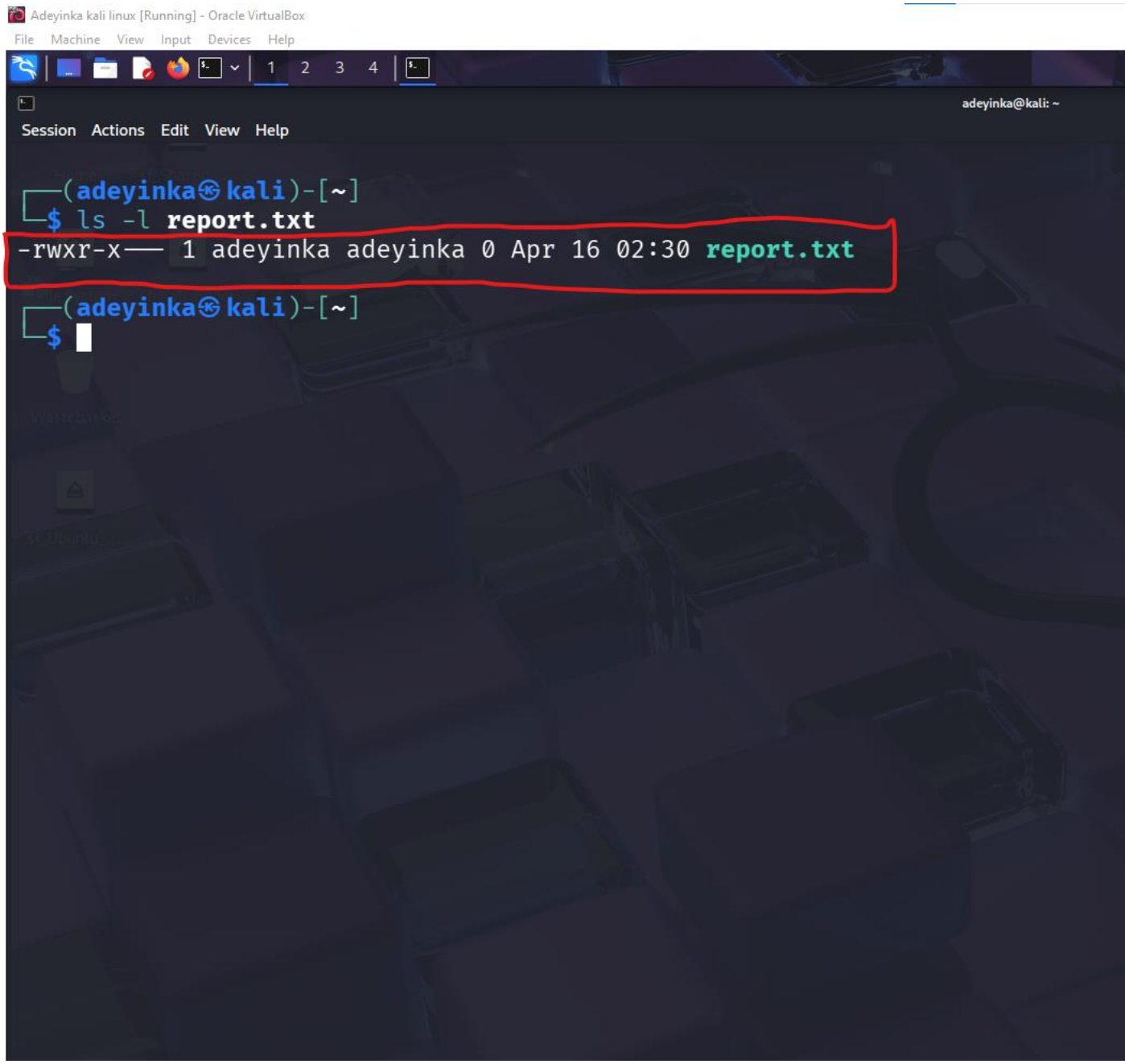
# Linux User Group & File Permission Management

**Prepare By:** **Martins Adeyanju** **Cybersecurity Analyst**

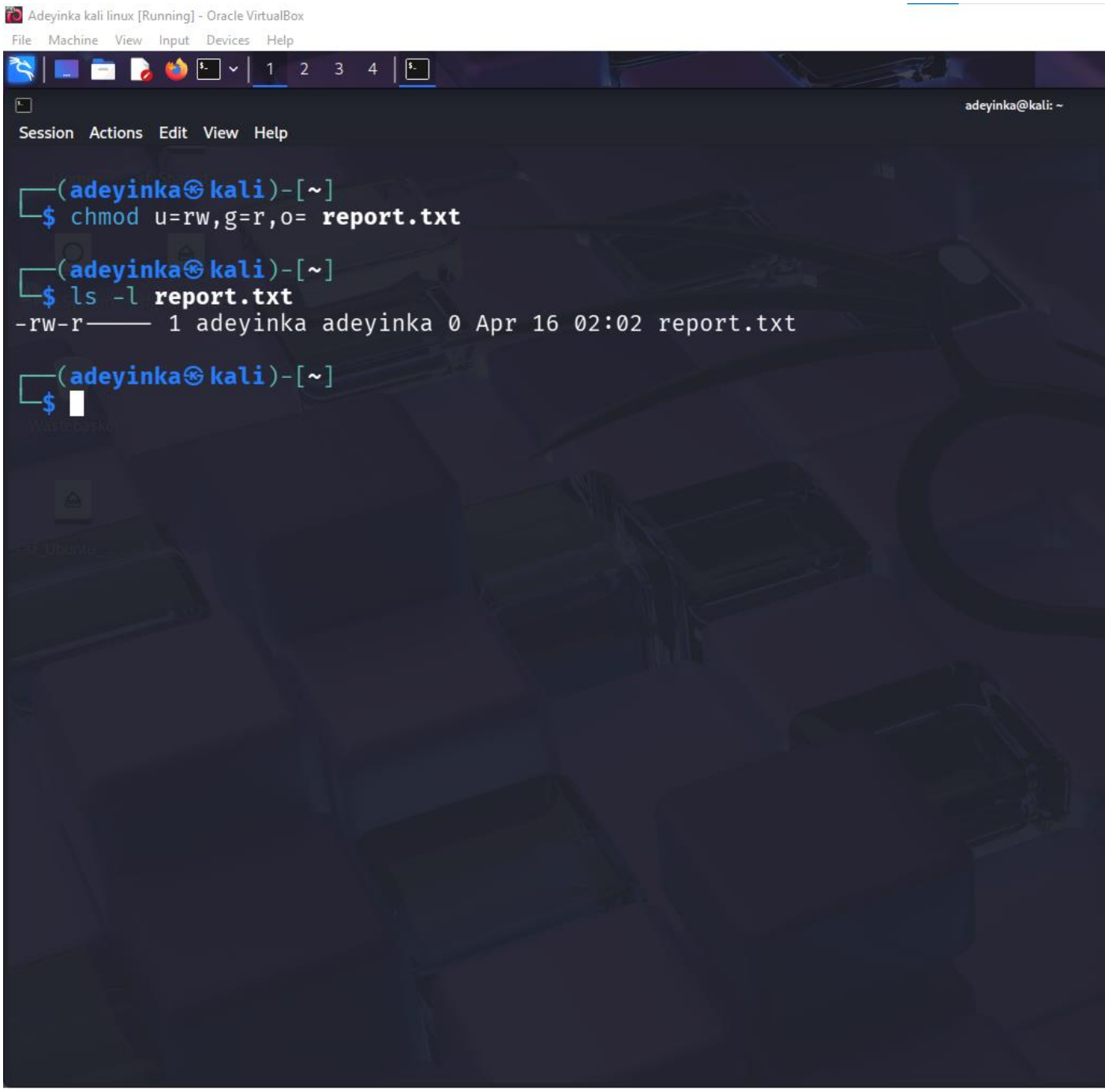
## CREATE FILE PERMISSION



## CURRENT PERMISSION VERIFICATION



**SYMBOLIC METHOD (RW-R-----)**




**NUMERIC METHOD (RWXR-X--- → 750)**

Adeyinka kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

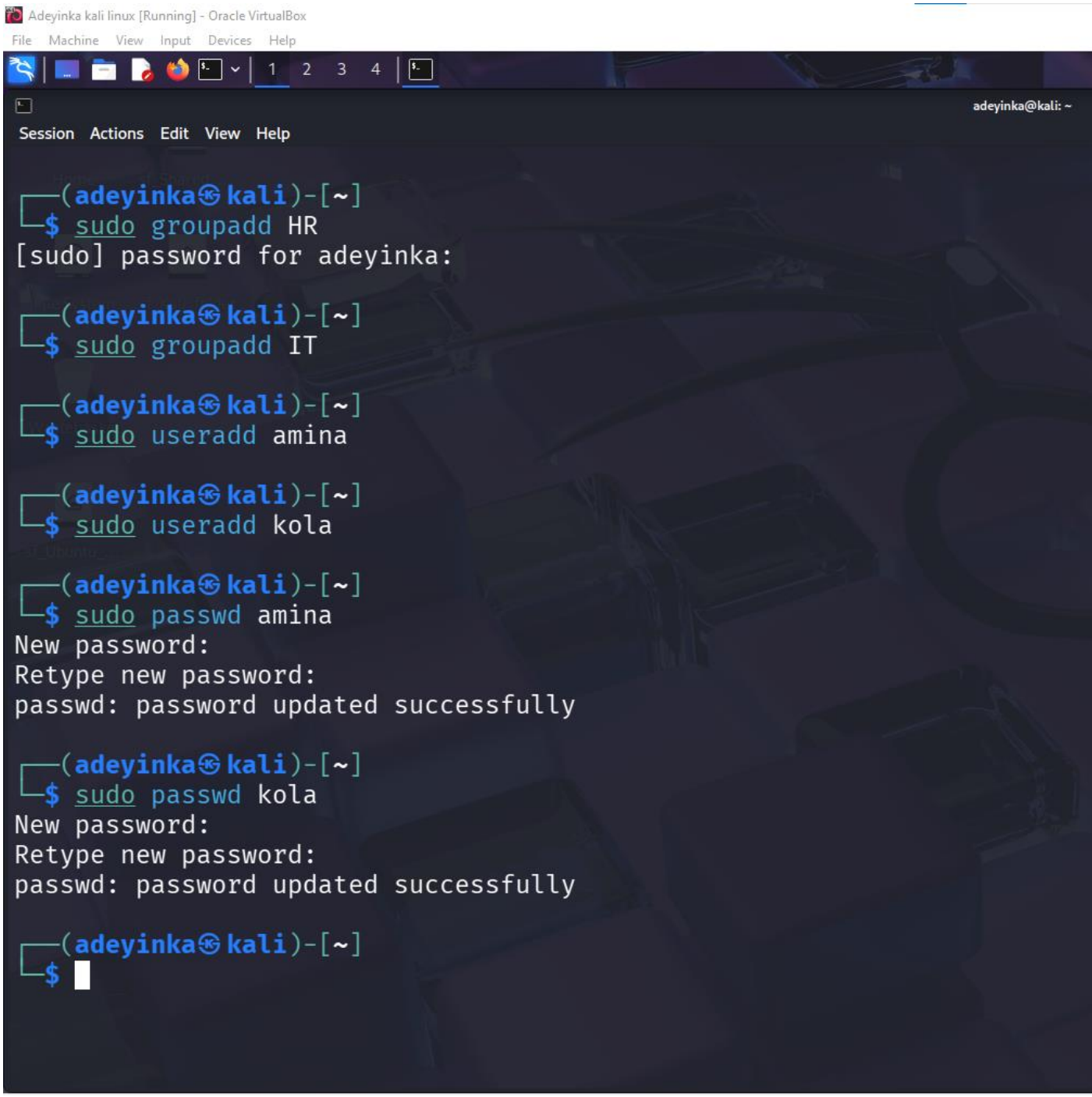
1 2 3 4

adeyinka@kali: ~

Session Actions Edit View Help

```
┌──(adeyinka㉿kali)-[~]
└─$ chmod 750 report.txt

┌──(adeyinka㉿kali)-[~]
└─$ ls -l report.txt
-rwxr-x── 1 adeyinka adeyinka 0 Apr 16 02:02 report.txt

┌──(adeyinka㉿kali)-[~]
└─$
```

## CREATE GROUPS, USERS AND SETTING PASSWORDS




## ADDING USERS TO GROUPS

```
┌──(adeyinka㉿kali)-[~]
└─$ sudo usermod -aG HR amina

┌──(adeyinka㉿kali)-[~]
└─$ sudo usermod -aG IT kola

┌──(adeyinka㉿kali)-[~]
└─$ id amina
uid=1001(amina) gid=1003(amina) groups=1003(amina),1001(HR)

┌──(adeyinka㉿kali)-[~]
└─$ id kola
uid=1002(kola) gid=1004(kola) groups=1004(kola),1002(IT)

┌──(adeyinka㉿kali)-[~]
└─$
```

**CREATE DIRECTORIES**

```
┌──(adeyinka㉿kali)-[~]
└─$ sudo mkdir -p /shared/hr_docs

┌──(adeyinka㉿kali)-[~]
└─$ sudo mkdir -p /shared/it_docs
```

**ASSIGN GROUP OWNERSHIP**

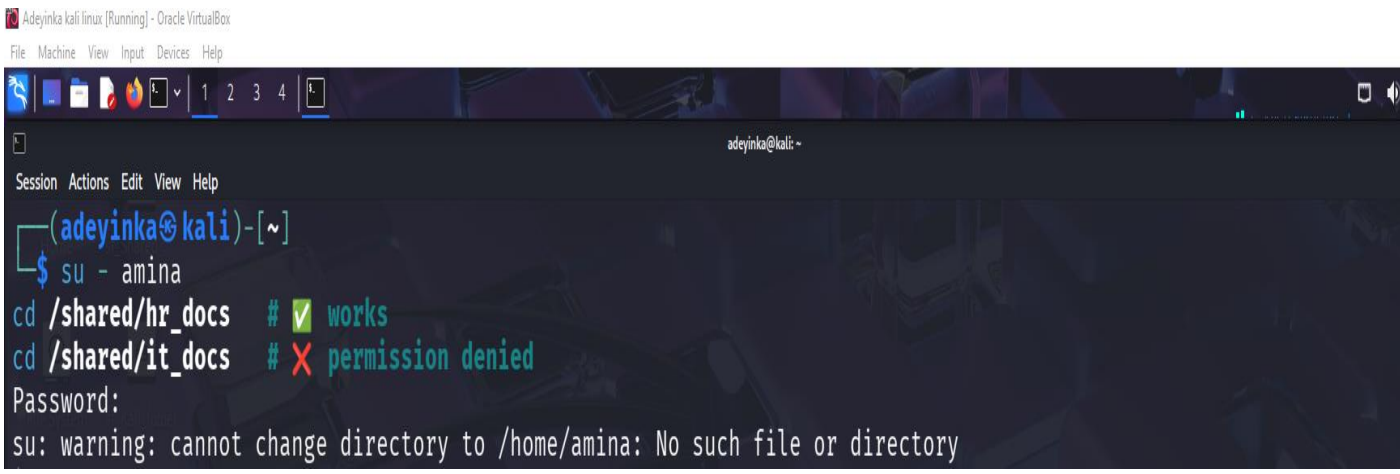
```
┌──(adeyinka㉿kali)-[~]
└─$ sudo chown :HR /shared/hr_docs

┌──(adeyinka㉿kali)-[~]
└─$ sudo chown :IT /shared/it_docs

┌──(adeyinka㉿kali)-[~]
└─$
```

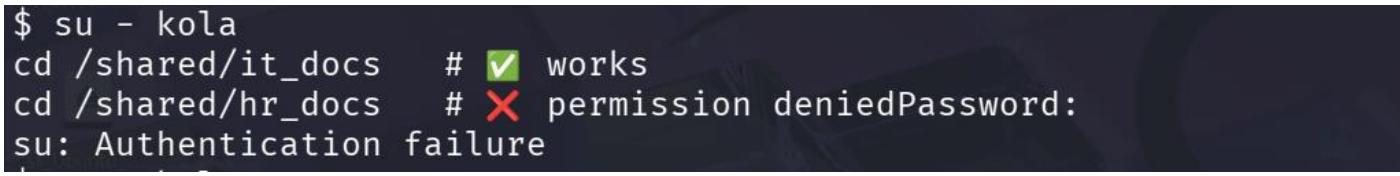
**ASSIGN GROUP OWNERSHIP AND SETTING PERMISSION**

```
┌──(adeyinka㉿kali)-[~]
└─$ sudo chmod 770 /shared/hr_docs

┌──(adeyinka㉿kali)-[~]
└─$ sudo chmod 770 /shared/it_docs

┌──(adeyinka㉿kali)-[~]
└─$ sudo touch /shared/hr_docs/hr_file.txt

┌──(adeyinka㉿kali)-[~]
└─$ sudo touch /shared/it_docs/it_file.txt

┌──(adeyinka㉿kali)-[~]
└─$ ls -ld /shared/*
drwxrwx--- 2 root HR 4096 Apr 16 03:32 /shared/hr_docs
drwxrwx--- 2 root IT 4096 Apr 16 03:32 /shared/it_docs

┌──(adeyinka㉿kali)-[~]
└─$
```

**ADD amina to IT group AND REMOVING amina from IT group**

```
┌──(adeyinka㉿kali)-[~]
└─$ sudo usermod -aG IT amina
id amina
uid=1001(amina) gid=1003(amina) groups=1003(amina),1001(HR),1002(IT)

┌──(adeyinka㉿kali)-[~]
└─$ sudo gpasswd -d amina IT
id amina
Removing user amina from group IT
uid=1001(amina) gid=1003(amina) groups=1003(amina),1001(HR)

┌──(adeyinka㉿kali)-[~]
└─$
```




## REFLECTION QUESTIONS

- What is the difference between the symbolic and numeric methods?

  **Answer:** Symbolic uses letters. i.e (User = rwx ) (Group = -wx) (Others = --x)
  Numeric uses numbers. i.e (User + Group + Others) = 731

- What does 640 mean? User = rw Group = r Others =none (rw-r-----)

- What does 750 mean? User = rwx Group = rx Others = none (rwxr-x---)

- Which method do you think is faster for an administrator, and why
  **Answer:** I think Numeric is faster due to its convenience